

Embedded Software Engineer

FF07 Software development and engineering · ISCO-08 2512 · CBS 0814 Software- en applicatieontwikkelaars
 · seniority medior · Also known as: Firmware Engineer, Embedded Developer, Embedded Systems Engineer

Develops software that runs inside devices and machines, taking hardware, timing and limited resources into account.

The embedded software engineer works at the boundary of electronics and software and delivers code that drives hardware directly in products or production equipment. Unlike an application developer, this role has no abundant resources and no easy update afterwards, so defects sit literally in the field. Selection targets language and sector, while the real criterion is whether someone can measure and debug systematically on a system that cannot be paused neatly, and can read a schematic and a datasheet.

Key figures for this profile

25 skills · 3 competencies to measure · 5 knockout skills · centre of gravity: Digital, data and AI 49%, Cognitive ability and problem solving 21%, Professional knowledge domains 12%

1 Competencies to measure

These are the behavioural and ability constructs in this profile. Each one shows which hrmforce questionnaire measures it.

Competencies to measure	Target level	hrmforce instrument	Assessment method	Behavioural anchor at target level
Accuracy and attention to detail Competency (50-framework): Nauwkeurigheid	 N5	Big Fifty, 360 Feedback	Persoonlijkheidsvragenlijst	Sets the standards for completeness and correctness, has samples checked and addresses teams on structural sloppiness.
Abstract and figural reasoning	 N4	Ability Scan	Cognitieve capaciteitentest	Discovers the underlying rule in unfamiliar material with several changing features and uses it to explain deviations.

Competencies to measure	Target level	hrmforce instrument	Assessment method	Behavioural anchor at target level
Working independently Competency (50-framework): Onafhankelijkheid	 N3	Big Fifty, 15PF, Competency Check	Persoonlijkheidsvragenlijst	Determines own approach for a full work package, consults colleagues purposefully when unclear and delivers on time without reminders.

2 Full skill profile

Every skill in this profile, sorted by weight. The behavioural anchor under each skill belongs to the target level.

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
G	Accuracy and attention to detail Sets the standards for completeness and correctness, has samples checked and addresses teams on structural sloppiness.	■■■■■ N5	5	no	Persoonlijkheidsvragenlijst	Big Fifty, 360 Feedback
V	Problem analysis Splits a recurring complaint into possible causes, tests these with own data and names the most likely cause.	■■■■■ N4	5	yes	Assessment center	Competency Check, Ability Scan
V	Programming Independently builds a function or module from the requirement, chooses suitable structures and keeps the code readable and tested.	■■■■■ N4	5	yes	Werkproef	Werkproef, Kennistoets (klantspecifiek)

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Software design and architecture Produces a design for a defined application, records the main choices and alternatives and reviews it with colleagues.	 N4	5	yes	Werkproef	Werkproef, Portfolio
A	Abstract and figural reasoning Discovers the underlying rule in unfamiliar material with several changing features and uses it to explain deviations.	 N4	4	no	Cognitieve capaciteitentest	Ability Scan
V	Modelling and abstraction Builds a diagram of a work process independently, names the assumptions and tests the model against real cases.	 N4	4	no	Werkproef	Werkproef, Portfolio

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
T	Version control and collaboration workflow Manages branches and merges independently, resolves conflicts and reviews colleagues' changes on substance.	 N4	4	yes	Werkproef	Werkproef
V	Software testing and quality assurance Designs test cases including edge cases, automates repeatable tests and advises on release based on the findings.	 N4	4	no	Werkproef	Werkproef, Kennistoets (klantspecifiek)
V	API integration Independently builds an interface between two systems, handles errors and documents the data exchange for support staff.	 N4	4	no	Werkproef	Werkproef

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
K	Technical understanding Independently explains the behaviour of an installation during a fault and traces the problem back to a technical principle.	 N4	4	no	Kennistoets	Ability Scan, Kennistoets (klantspecifiek)
V	Secure development Recognises vulnerabilities in design and code, resolves them and includes security tests in own way of working.	 N3	4	yes	Werkproef	Werkproef, Kennistoets (klantspecifiek)
V	Defining and managing requirements Elicits requirements from various stakeholders, makes them testable, prioritises together with them and manages changes traceably.	 N3	4	no	Werkproef	Werkproef, Structureerd interview

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	multidisciplinary collaboration Maps the issue with specialists from different disciplines and formulates a joint approach with a clear division of roles.	 N3	3	no	Assessment center	360 Feedback, Competency Check
V	Working in projects Draws up project plan, schedule and risk list for a medium sized project and reports periodically to the sponsor.	 N3	3	no	Werkproef	Kennistoets (klantspecifiek), Werkproef, Competency Check
G	Working independently Determines own approach for a full work package, consults colleagues purposefully when unclear and delivers on time without reminders.	 N3	3	no	Persoonlijkheidsvragenlijst	Big Fifty, 15PF, Competency Check

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Professional currency and specialist literature Tracks sources and guidelines structurally, weighs their quality and adapts the own working method to new insights.	 N3	3	no	Portfolio/bewijsstuk	Portfolio, Kennistoets (klantspecifiek)
V	Writing technical documentation Writes complete documentation for a system or process, tests it with an unfamiliar reader and maintains versions.	 N3	3	no	Werkproef	Werkproef, Portfolio
V	Release management and deployment Draws up a rollout plan with a fallback option, aligns with stakeholders and verifies after rollout that the service works.	 N3	3	no	Werkproef	Kennistoets (klantspecifiek), Werkproef

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Prompting and task delegation to AI Splits a work task into steps, decides which step AI does and which not and improves the instruction based on outcomes.	 N3	3	no	Werkproef	Werkproef, Portfolio
V	Electrical engineering work Independently locates a fault in a control system, measures voltage and continuity and repairs it within the safety standard.	 N3	3	no	Certificaat/diploma	Kennistoets (klantspecifiek), Werkproef
V	Measurement and calibration Independently selects the right instrument for a tolerance requirement, weighs measurement uncertainty and passes or rejects the product.	 N3	3	no	Werkproef	Werkproef, Kennistoets (klantspecifiek)

T	Skill	Target level	Weight	Knockout	Assessment method	hrmforce instrument
V	Written expression Independently writes texts with opening, core and closing, chooses a tone per audience and applies fixed terms consistently.	 N3	2	no	Werkproef	Werkproef, Competency Check
V	Evaluating and verifying AI output Determines per task which checks are needed, spots incorrect or fabricated parts and accounts for the result used.	 N3	2	no	Werkproef	Werkproef, Kennistoets (klantspecifiek)
V	giving feedback Discusses unwanted behaviour with the person concerned, describes fact, effect and request, and agrees on a follow-up.	 N3	1	no	Assessment center	360 Feedback, Communicatiestijlen
V	Using cloud platforms Uses existing cloud services as instructed and reports unusual behaviour or unexpected costs to the administrator.	 N2	1	no	Kennistoets	Kennistoets (klantspecifiek)

3 Levels per seniority

The same profile with the job family seniority adjustment applied. Target levels are clamped to 1 to 5.

Skill	junior	medior	senior
Accuracy and attention to detail	N5	N5	N5
Problem analysis	N3	N4	N5
Programming	N4	N4	N4
Software design and architecture	N3	N4	N5
Abstract and figural reasoning	N3	N4	N5
Modelling and abstraction	N3	N4	N5
Version control and collaboration workflow	N3	N4	N5
Software testing and quality assurance	N3	N4	N5
API integration	N3	N4	N5
Technical understanding	N3	N4	N5
Secure development	N2	N3	N4
Defining and managing requirements	N2	N3	N4
multidisciplinary collaboration	N2	N3	N4
Working in projects	N2	N3	N4
Working independently	N2	N3	N4
Professional currency and specialist literature	N2	N3	N4
Writing technical documentation	N2	N3	N4
Release management and deployment	N2	N3	N4
Prompting and task delegation to AI	N2	N3	N4
Electrical engineering work	N2	N3	N4
Measurement and calibration	N2	N3	N4
Written expression	N2	N3	N4
Evaluating and verifying AI output	N2	N3	N4
giving feedback	N2	N3	N4
Using cloud platforms	N2	N2	N2

4 The proficiency scale

Level	Label	Autonomy	Complexity	Scope	Depth of knowledge
N1	Guided	works under supervision and follows instruction	routine, one variable at a time	own task	basic understanding of terminology
N2	Applying	works independently within set frameworks	familiar situations with limited variation	own role	working knowledge, applies standard approach
N3	Proficient	sets own approach and seeks input proactively	several variables, some ambiguity	own team or process	thorough knowledge, weighs alternatives
N4	Advanced	directs others and decides on the approach	complex, with conflicting interests	several teams or a department	in-depth knowledge, advises others
N5	Leading	sets the standard and the policy	strategic, under high uncertainty	organisation, value chain or profession	authority, advances the profession

5 Behavioural anchors per skill

Anchors sit at N1, N3 and N5. N2 and N4 are deliberately not anchored: raters interpolate between them, following the O*NET convention.

Skill	N1	N3	N5
Accuracy and attention to detail Works carefully, checks own work for errors and keeps to agreements about completeness and correctness of data.	Goes through the own work with a checklist and repairs errors found before the work is passed on.	Builds check moments into the own process, keeps track of errors and prevents repetition by adjusting the method.	Sets the standards for completeness and correctness, has samples checked and addresses teams on structural sloppiness.
Problem analysis Breaks a problem into parts, separates causes from effects and states which information is needed to move forward.	States precisely what deviates in a malfunction and which data are missing, and puts this to the supervisor.	Splits a recurring complaint into possible causes, tests these with own data and names the most likely cause.	Traces persistent organisational problems back to a few root causes and sets the analysis method that other teams will use.
Programming Writes, reads and improves working program code in at least one language, splits a problem into steps and tests own solution.	Adjusts existing code as directed, runs it and reports error messages that cannot be resolved alone.	Independently builds a function or module from the requirement, chooses suitable structures and keeps the code readable and tested.	Sets programming standards and language choices for the organisation and reviews others' code for quality and maintainability.

Skill	N1	N3	N5
Software design and architecture Designs the structure of an application or interface, distributes functionality across components and weighs choices on maintainability, performance and security.	Elaborates a supplied design into components and asks questions when the design turns out to be incomplete.	Produces a design for a defined application, records the main choices and alternatives and reviews it with colleagues.	Sets the organisation's architecture principles and decides on designs that affect multiple systems.
Abstract and figural reasoning Recognises rules and patterns in figures, series or symbols without prior knowledge and applies those rules to new, unfamiliar tasks.	Finds the next step in a short series of figures and explains which rule was used for it.	Discovers the underlying rule in unfamiliar material with several changing features and uses it to explain deviations.	Unravels problems without a known example, formulates a new ordering and shapes it so that peers can build on it.
Modelling and abstraction Turns a real situation into a simplified representation, such as a diagram, formula or flow chart, with explicit assumptions and limits.	Fills in an existing diagram or template and names which part of the work belongs where.	Builds a diagram of a work process independently, names the assumptions and tests the model against real cases.	Develops models that several departments use, states their limits of validity and manages their versions and changes.
Version control and collaboration workflow Works with a version control system, uses branches and merge requests and keeps the change history traceable for the team.	Commits changes with a clear description in the designated branch according to the team agreement.	Manages branches and merges independently, resolves conflicts and reviews colleagues' changes on substance.	Determines the way of working for version control and code review in the organisation and enforces it across teams.
Software testing and quality assurance Devises and runs tests on functionality and edge cases, records findings reproducibly and judges whether a solution can be released.	Works through a test script step by step and records every deviation with a screenshot and the steps taken.	Designs test cases including edge cases, automates repeatable tests and advises on release based on the findings.	Sets the organisation's test strategy and quality requirements and decides when software is not released.
API integration Connects systems through programming interfaces, reads their documentation, handles errors and limits and tests the interface end to end.	Executes a prepared interface call following documentation and reports recurring error codes to the developer.	Independently builds an interface between two systems, handles errors and documents the data exchange for support staff.	Sets the organisation's integration standards and interface policy and decides on exposure to external parties.

Skill	N1	N3	N5
Technical understanding Understands how machines, structures and installations work, recognises causes of failure and explains them using technical and physical principles.	Names the parts of a simple machine and explains, using the manual, what function each part has.	Independently explains the behaviour of an installation during a fault and traces the problem back to a technical principle.	Develops technical guidelines for the field and is consulted internally and externally on design and failure questions.
Secure development Builds software in which input is validated, secrets are protected and known vulnerabilities are prevented, and verifies this with targeted tests.	Follows the secure coding guideline for own changes and asks when a guideline is unclear.	Recognises vulnerabilities in design and code, resolves them and includes security tests in own way of working.	Sets secure development requirements in the organisation and judges whether solutions meet them before release.
Defining and managing requirements Elicits requirements from users and clients, records them testably, tracks changes and checks whether the result meets them.	Records user needs in the agreed format and probes until every need is concrete.	Elicits requirements from various stakeholders, makes them testable, prioritises together with them and manages changes traceably.	Determines the requirements practice in the organisation and decides which requirements prevail when interests conflict.
multidisciplinary collaboration Working with people from other disciplines by learning their language and methods and establishing shared starting points for the joint work.	Asks a colleague from another discipline what technical terms mean and summarises the answer back to them.	Maps the issue with specialists from different disciplines and formulates a joint approach with a clear division of roles.	Sets up lasting forms of collaboration between disciplines and settles conflicts between professional standards on the basis of the organisational interest.
Working in projects Delivering a defined result through phases, milestones and role division, controlling time, money, quality and risks and reporting at fixed moments.	Delivers assigned project tasks on time and updates the project plan with own progress.	Draws up project plan, schedule and risk list for a medium sized project and reports periodically to the sponsor.	Establishes the project approach of the organisation, decides on starting and stopping programmes and safeguards coherence between projects.
Working independently Carries out assigned work without ongoing direction, makes own choices within agreed boundaries and reports when consultation or a decision by others is needed.	Completes tasks as instructed and raises every deviation with the supervisor before continuing with the remaining work.	Determines own approach for a full work package, consults colleagues purposefully when unclear and delivers on time without reminders.	Designs the working method that lets teams operate autonomously and defines across the organisation which decisions are escalated and which are not.

Skill	N1	N3	N5
Professional currency and specialist literature Follows developments in the own profession through literature, guidelines and networks and works relevant new insights into own practice.	Reads the professional information received and names which part matters for the own work.	Tracks sources and guidelines structurally, weighs their quality and adapts the own working method to new insights.	Assesses the state of the field, weighs conflicting research and determines which guidelines the organisation will follow from now on.
Writing technical documentation Writes manuals, support documentation and decision records that a reader without prior knowledge can follow step by step and that stay current.	Completes a documentation template with the steps of a task and has the text checked by a colleague.	Writes complete documentation for a system or process, tests it with an unfamiliar reader and maintains versions.	Sets the organisation's documentation standard and ensures documentation is part of every delivery.
Release management and deployment Plans and carries out the rollout of new versions, checks operation before and after and ensures a rollback stays possible.	Executes the steps of a rollout plan and reports immediately when a step does not give the expected result.	Draws up a rollout plan with a fallback option, aligns with stakeholders and verifies after rollout that the service works.	Sets the organisation's release policy, decides on rollout windows and structurally improves the lead time of changes.
Prompting and task delegation to AI Divides work between person and AI system, formulates task, context and required output format and adjusts based on the output.	Gives an AI tool a single task with a clear question and required format and uses the result after checking it.	Splits a work task into steps, decides which step AI does and which not and improves the instruction based on outcomes.	Develops organisation wide practices for dividing tasks with AI and trains others in effective and responsible instruction giving.
Electrical engineering work Installs, measures and repairs electrical installations and control systems according to diagrams and safety rules, and makes the installation safe to work on.	Installs electrical components under supervision according to the diagram and uses the prescribed protective equipment.	Independently locates a fault in a control system, measures voltage and continuity and repairs it within the safety standard.	Draws up the company's electrical work instructions and safety rules and reviews installation designs for compliance with standards.
Measurement and calibration Takes measurements with measuring equipment, judges whether results fall within tolerance and keeps equipment calibrated and traceable.	Measures with calliper and dial gauge under supervision and notes the result on the measurement form.	Independently selects the right instrument for a tolerance requirement, weighs measurement uncertainty and passes or rejects the product.	Sets up the organisation's calibration management, writes measurement instructions and accounts for traceability during external audits.

Skill	N1	N3	N5
Written expression Putting thoughts into correct and readable text with a logical paragraph structure, fitting tone and word choice suited to the reader.	Writes short messages and reports free of language errors after using a checking tool and a review round by a colleague.	Independently writes texts with opening, core and closing, chooses a tone per audience and applies fixed terms consistently.	Draws up the writing guide and house style for the organisation, approves publications on content and develops the writing skill of teams.
Evaluating and verifying AI output Checks AI output for accuracy, completeness and usefulness, verifies claims against sources and decides whether the output is used.	Checks facts and figures from AI output against a reliable source before passing the result on.	Determines per task which checks are needed, spots incorrect or fabricated parts and accounts for the result used.	Designs the organisation's verification policy and determines for which applications human review remains mandatory.
giving feedback Naming concrete behaviour and its effect to another person, with a request or proposal, at a moment and in a way that keeps dialogue possible.	States after a shared task what a colleague did well and gives a concrete example with it.	Discusses unwanted behaviour with the person concerned, describes fact, effect and request, and agrees on a follow-up.	Introduces feedback as a working method across several teams, trains managers in it and monitors whether unwanted behaviour is raised earlier.
Using cloud platforms Sets up and manages services in a cloud environment, arranges permissions and costs and chooses between available service types based on requirements.	Uses existing cloud services as instructed and reports unusual behaviour or unexpected costs to the administrator.	Independently configures cloud services for an application, arranges permissions and backups and keeps costs and usage in view.	Determines the organisation's cloud strategy and decides which workloads are placed where.

6 Assessment methods and validity

The validities are the corrected meta-analytic estimates from Sackett, Zhang, Berry and Lievens (2022) in the Journal of Applied Psychology, which replace the older Schmidt and Hunter (1998) figures. The spread matters as much as the mean: with a high SD, validity varies strongly by context.

Assessment method	rho	SD	What it is
Persoonlijkheidsvragenlijst	0.40	0.15	Self-report of behavioural preferences. At hrmforce the Big Fifty and the 15PF.
Assessment center	0.29	0.09	Multiple exercises, multiple assessors, dimension-wise scoring.
Werkproef	0.33	0.09	The candidate performs a representative work sample under standardised conditions.
Cognitieve capaciteitentest	0.41	0.14	Standardised test of verbal, numerical or abstract reasoning.
Kennistoets	0.40	0.13	Test of declarative job knowledge, usually assembled per client.
Portfolio/bewijsstuk	-	-	A collection of prior work or results, assessed against fixed criteria.
Certificaat/diploma	-	-	Formal proof of completed education or examination.

7 Assessment instruments for this job

hrmforce instrument	Skills	What it measures
Werkproef	16	Representatief werkfragment onder gestandaardiseerde condities
Kennistoets (klantspecifiek)	11	Toets van vakkennis, per klant samengesteld
Competency Check	5	Competentiebeoordeling op gedragsniveau
Portfolio	5	Beoordeling van eerder werk tegen vaste criteria
360 Feedback	3	Meervoudige beoordeling op gedragsankers
Ability Scan	3	Capaciteitentest: verbaal, numeriek en abstract redeneervermogen
Big Fifty	2	Persoonlijkheidsvragenlijst, 150 items, 25 factoren
Structureerd interview	1	Gedragsgericht interview met vaste opzet
15PF	1	Persoonlijkheidsprofiel, korter alternatief voor de Big Fifty
Communicatiestijlen	1	Voorkeursstijl in communicatie

8 How to use this profile

1. Agree the profile with the hiring manager before the first candidate is measured. Weights shifted afterwards make the outcome indefensible.
2. Choose the assessment method per skill from the Assessment method column. Combine at most three methods per candidate.
3. Calibrate the assessors in a two-hour session using the anchors in chapter 5. Anchors without rater training deliver little.
4. Fill in the match calculator and discuss the gaps in the development conversation. Type A and type G are selection criteria, not one-quarter development goals.

Source: hrmforce Skills Framework 2.0, version 2.0.0, built 2026-09-08. Crosswalks to ESCO v1.2.1, O*NET 31.0, Lightcast Open Skills, SFIA 9, DigComp 3.0, EntreComp, LifeComp, GreenComp, AILit, WEF Future of Jobs 2025, ISCO-08 and the Dutch CBS occupational classification BRC 2014 edition 2025. This profile is a starting point, not a standard: align it with your own organisation before selecting on it.